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Hom-Functors and Representable Functors

Hom-Functors

Covariant Hom $_C(A, -)$

Functor $C \rightarrow \text{Set}$

$X \mapsto \text{Hom}_C(A, X)$ [morphisms FROM A]
 $(f: X \rightarrow Y) \mapsto (f \circ -): \text{Hom}(A, X) \rightarrow \text{Hom}(A, Y)$

Functoriality: Composition preserved

Contravariant Hom $_C(-, A)$

Functor $C^{\text{op}} \rightarrow \text{Set}$

$X \mapsto \text{Hom}_C(X, A)$ [morphisms TO A]
 $(f: X \rightarrow Y) \mapsto (- \circ f): \text{Hom}(Y, A) \rightarrow \text{Hom}(X, A)$

Arrows reversed due to contravariance

Representable Functors

Definition: $F: C \rightarrow \text{Set}$ is **representable** if $\exists A \in C$ such that:

$F \cong \text{Hom}_C(A, -)$ [natural isomorphism]

Universal Element: Unique $a \in F(A)$ such that $F(f)(a) = (f,a)$ establishes bijection

Criterion for Representability

Brown Representability: F is representable iff: 1. Preserves limits 2. Solution set condition holds

10 Detailed Examples

1. Forgetful Functor $\text{Grp} \rightarrow \text{Set}$

$U(G)$ = underlying set of G
 $\text{Hom}_{\text{Grp}}(\mathbb{Z}, G) \cong U(G)$ [homomorphisms = elements]
 Represented by $\mathbb{Z}$

2. Polynomial Evaluation

$F: \text{Ring} \rightarrow \text{Set}, R \mapsto R^n$ [n -tuples]
 $\text{Hom}_{\text{Ring}}(\mathbb{Z}[x_1, \dots, x_n], R) \cong R^n$ [evaluation]
 Represented by polynomial ring

3. Continuous Functions to Sierpinski

$F: \text{Top} \rightarrow \text{Set}, X \mapsto \text{open sets of } X$
 $\text{Hom}(X, \text{Sierpinski}) \cong \text{Open}(X)$ [continuous = inclusion]
 Represented by Sierpinski space

4. Power Set Functor

$F: \text{Set} \rightarrow \text{Set}, X \mapsto P(X)$
 $\text{Hom}(X, \{0,1\}) \cong P(X)$ [characteristic functions]
 Represented by two-element set

5. Tangent Bundle (Differential Geometry)

$T: \text{Diff} \rightarrow \text{Set}, M \mapsto \text{tangent bundle}$
 Representable by infinitesimal construction

6. Fundamental Group (Pointed Spaces)

$\pi_1: \text{Top}_* \rightarrow \text{Grp}$
 $\text{Hom}(S^1, X) \cong \pi_1(X)$ [loops = elements]
 Represented by S^1

7. Cohomology Ring (Homotopy)

$H^*: \text{Top} \rightarrow \text{Grp}$
 Representable by Eilenberg–MacLane spaces

8. Tensor Algebra

$T: \text{Vect} \rightarrow \text{Vect}, V \mapsto \otimes^* V$
 $\text{Hom}(T(V), W) \leftrightarrow \text{morphisms respecting tensor structure}$

9. Free Module Functor

$F: \text{Grp} \rightarrow \text{Set}, X \mapsto \text{free } \mathbb{Z}[G]\text{-module on } X$
 $\text{Hom}(\mathbb{Z}[G], M) \cong M^G$ [G-module structure]

10. Moduli Space

Schemes parameterized by objects
 Hilbert scheme: $\text{Hom}(-, H)$ represents closed subschemes

Yoneda Embedding

Embedding $y: C \rightarrow [C^{\text{op}}, \text{Set}]$

$y(A) = \text{Hom}_C(-, A)$
 $y(f: A \rightarrow B) = \text{Hom}(f, -): \text{Hom}(-, A) \Rightarrow \text{Hom}(-, B)$

Theorem: y is fully faithful

Consequence: Every category embeds in presheaves; presheaves are free cocompletions

Power of Representability

1. **Universal Property Characterization:** Universal properties = representability statements
2. **Computational Power:** Categorical $\rightarrow$ set-theoretic (via representation)
3. **Automatic Existence/Uniqueness:** From representing object
4. **Natural Constructions:** Representable functors are “canonical”
5. **Foundation for Adjoints:** Adjoints define themselves via representability

Adjunctions and Representability

Fundamental Connection:

$F \dashv G \iff \text{Hom}_D(F(c), d) \cong \text{Hom}_C(c, G(d))$ [natural in both variables]

Right adjoint G : “Represents” the hom-functor

Common Mistakes

- ✗ “All functors are representable” ✓ Only special ones; most are not
- ✗ “Representability = free functor” ✓ Related but different concepts
- ✗ “Solution set condition not important” ✓ Necessary for representability to exist

Study Guide

1. Master $\text{Hom}_C(A, -)$ definition
2. Understand contravariance in $\text{Hom}_C(-, A)$
3. Work through 5+ examples

4. Prove Yoneda embedding fully faithful
5. Connect to adjoint functors
6. Apply representability to your field

Key Questions

1. Why is $\text{Hom}_C(A, -)$ a functor?
2. What makes a functor representable?
3. Why is Yoneda embedding important?
4. How do represent functors relate to adjoints?
5. What's the geometric meaning of representable?